

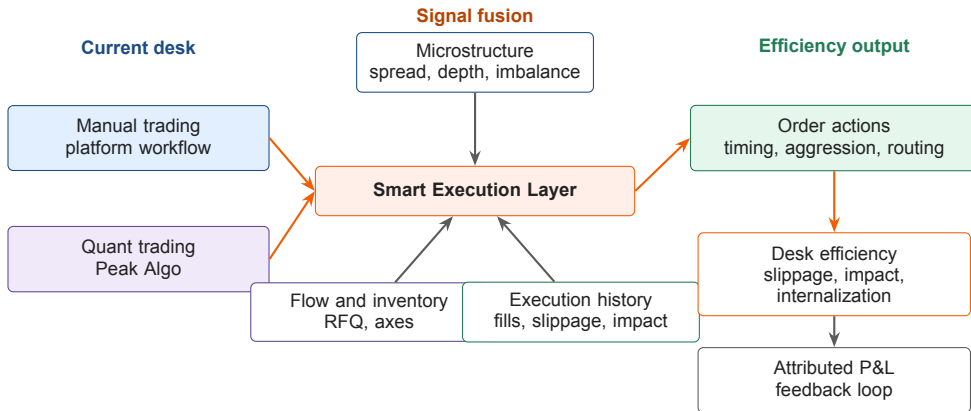
# Liquidity-Aware Smart Execution for Desk Efficiency

Microstructure Signals Across Manual Trading and Peak Algo

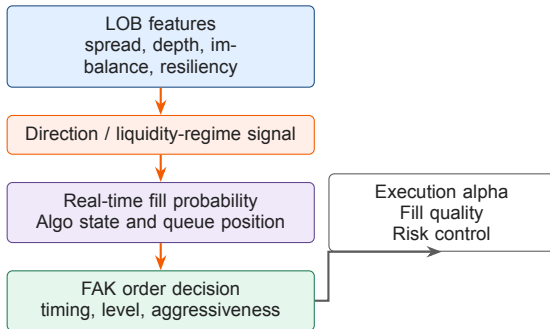
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## ■ Smart Execution as a Desk Efficiency Layer

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# ■ FAK Order Execution Alpha



## Case 1: favorable direction

- ▶ Direction supports the order side.
- ▶ Expected fill probability is higher.
- ▶ Execution alpha: lower slippage and lower market impact.

## Case 2: adverse direction

- ▶ Direction works against the FAK order.
- ▶ Revise fill expectation and aggressiveness.
- ▶ Improve success rate and reduce unhedged risk.

## ■ Algo Plan and Literature Support

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### Implementation plan

1. Build LOB feature stream by execution horizon and liquidity regime.
2. Start with transparent features: spread, depth, imbalance, queue depletion, resiliency.
3. Join direction signal with Algo fill-probability engine.
4. Run shadow trading, then controlled A/B test.

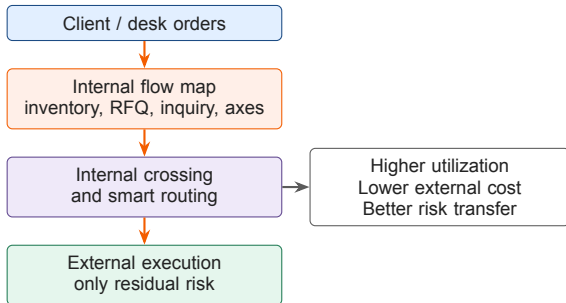
### Why feasible

- ▶ Queue models link best-queue depletion to short-term price moves.
- ▶ Hawkes-style order-flow models support regime-dependent intensity signals.
- ▶ DeepLOB-style evidence shows book snapshots contain short-horizon structure.
- ▶ Microstructure benchmarks warn us to evaluate by cost-aware execution metrics, not accuracy alone.

Source: Cont and de Larrard; Bacry et al.; Zhang et al. DeepLOB; recent LOB benchmark studies.

## ■ Microstructure for Internal Flow

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### How microstructure helps

- ▶ Identify when external liquidity is expensive.
- ▶ Prioritize internal crossing before paying spread and impact.
- ▶ Route residual flow when book depth and resiliency are favorable.
- ▶ Feed realized cost back into desk-level flow intelligence.

# ■ How We Save 0.01 bp Per Trade

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## Base-case value

**90M**

annual P&L target from 0.01 bp saving

- ▶ 2025 flow: 45tn.
- ▶ Average duration: 2 years.
- ▶  $45tn \times 2 \times 0.01bp \approx 90M$ .

## Where the saving comes from

- ▶ **Internal crossing:** avoid spread and external market impact.
- ▶ **Timing:** trade when depth and resiliency are favorable.
- ▶ **Adverse-selection control:** avoid passive fills when direction is unfavorable.

## How we prove it

- ▶ Slippage vs. arrival / decision price.
- ▶ Market impact after execution.
- ▶ Internalization uplift and residual external cost.